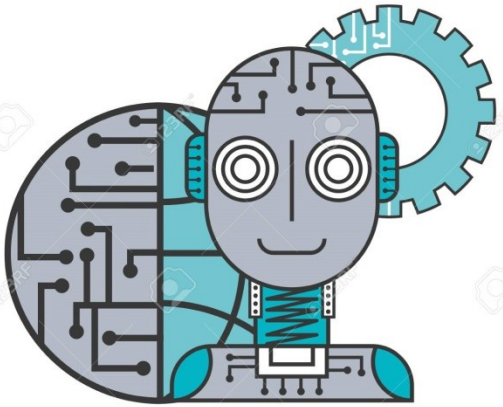
Module 6CM521: Frontiers in AI and Data Science



Data Science: data preprocessing

Ioannis Tsioulis (MC)

Slides acknowledgements:
Dr Hong Qing Yu

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Declaration

The lecture slides are based on the book: Data Ming, concepts and technologies, *Chapter 3*

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Agenda

Why data preprocessing?

Data cleaning

Data integration and transformation

Data reduction

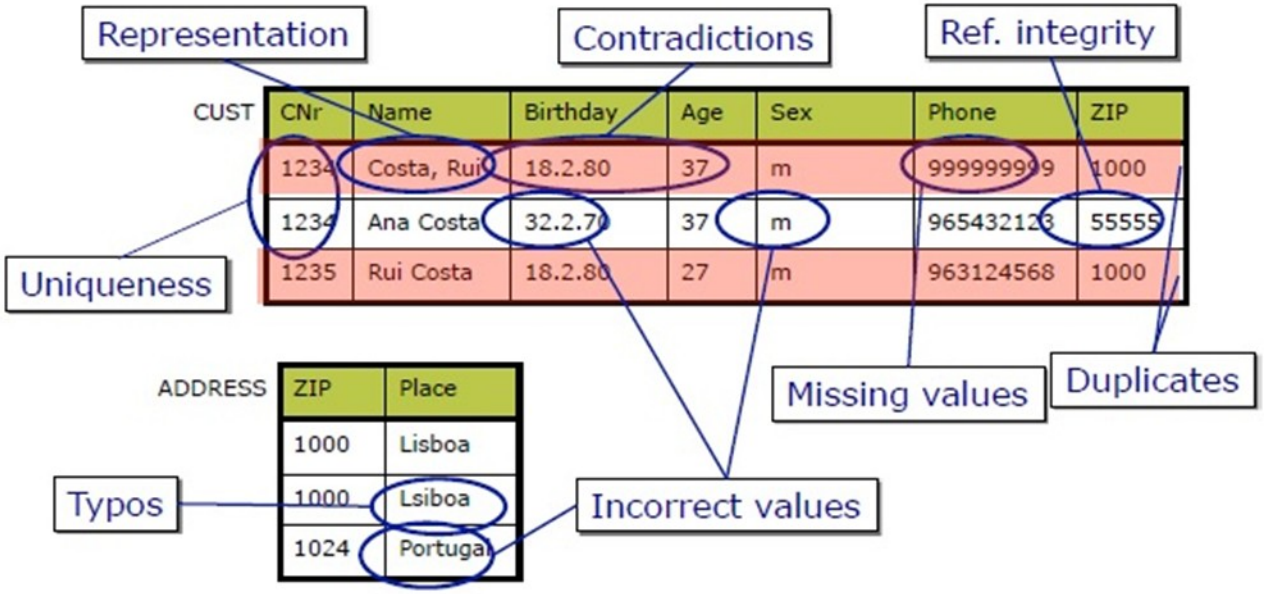
Summary

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Why Data Preprocessing?

Data in the real world is dirty! Missing, noisy and inconsistent data.



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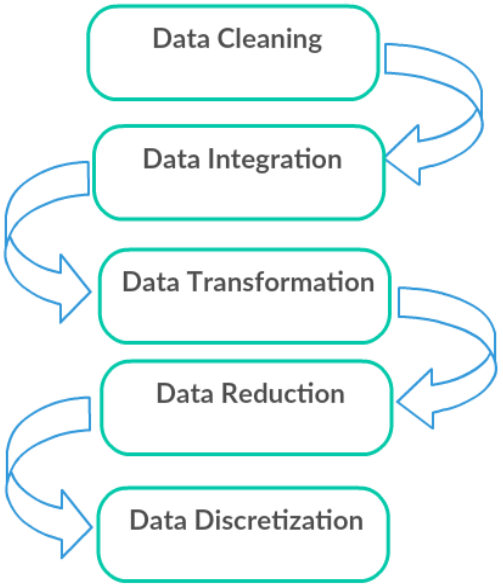
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Major Tasks in Data Preprocessing

- Data cleaning
  - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- Data integration
  - Integration of multiple databases, data cubes, files, or notes
- Data transformation
  - Normalization (scaling to a specific range)
  - Aggregation
- Data reduction
  - Obtains reduced representation in volume but produces the same or similar analytical results
- Data discretization
  - With particular importance, especially for numerical data
  - We will discuss this in detail later



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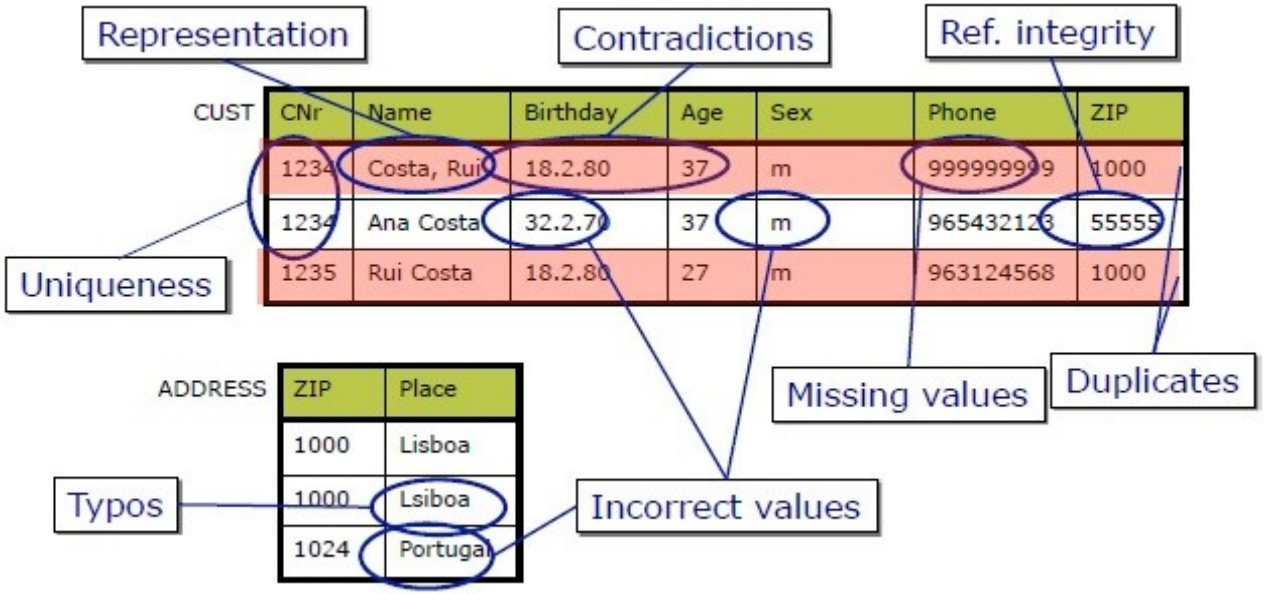
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Data Cleaning

Data cleaning tasks

- Fill in missing values
- Identify outliers and smooth out noisy data
- Correct inconsistent data



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Missing Data

Data is not always available

- E.g., many tuples have no recorded value for several attributes, such as customer income in sales data

Missing data may be due to

- equipment malfunction
- inconsistent with other recorded data and thus deleted
- data not entered due to misunderstanding
- certain data may not be considered important at the time of entry
- not register history or changes of the data

Missing data may need to be inferred

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Missing Data Example

Name	Age	Height	Weight	BMI	Symptom severity	Medicine type	Recovery time
Emma	55.5	159	60	23.7	NA	B	13
Isabella	NA	164	65	24.2	8	B	NA
Liam	37.2	186	NA	NA	NA	A	5
Evelyn	NA	156	56	23.0	4	B	15
Joey	37.0	188	87	24.6	NA	A	5
Charlotte	46.5	166	NA	NA	2	A	8
Logan	42.1	186	83	24.0	5	B	NA

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How to Handle Missing Data?

Ignore the tuple: usually done when class label is missing (assuming the task is classification—not effective in certain cases)

Fill in the missing value manually: tedious + infeasible?

Use a global constant to fill in the missing value: e.g., “unknown”, a new class?!

Use the attribute mean to fill in the missing value

Use the attribute mean for all samples of the same class to fill in the missing value: smarter

Use the most probable value to fill in the missing value: inference-based such as regression, Bayesian formula, decision tree

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Difference kinds of missing data

- **Data missing completely at random (MCAR)**: missingness does not depend on any values of the missing or measured data. That is missingness r_i does not depend on the (unobserved) value y_i or on observed values x_i. In this case, entities with missing data can be removed from the analysis safely.

- **Data missing at random (MAR)**: missingness ri does not depend on value of y_i, but may depend on the value of x_i. Here, removing data can bias analysis since you would drop values of x based on missingness and potentially change the distribution of x.

- **Data not missing at random (NMAR)**: missingness r_i depends on y_i. This is the most pernicious of all, and usually means that we want to go back to our collaborator and tell them that we are in a bind. Removing or imputing data as discussed is not appropriate in this case.

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Result

	ID	Age	BloodPressure
0	1	33	135.0
1	2	36	NaN
2	3	55	119.0
3	4	59	145.0
4	5	23	123.0
5	6	21	140.0
6	7	25	124.0
7	8	23	NaN
8	9	48	NaN
9	10	37	NaN

	ID	Income	EducationYears
0	1	NaN	8
1	2	NaN	11
2	3	42361.0	11
3	4	76873.0	12
4	5	43419.0	14
5	6	NaN	
6	7	77010.0	
7	8	NaN	
8	9	NaN	
9	10	71942.0	

	ID	Age	BMI	ReportedCalories	ExerciseHoursPerWeek
0	1	45	26.2	NaN	6
1	2	42	29.8	2141.0	3
2	3	56	22.6	NaN	6
3	4	50	22.2	NaN	7
4	5	18	20.9	3767.0	5
5	6	44	21.7	3260.0	5
6	7	69	27.5	NaN	7
7	8	30	24.9	3259.0	4
8	9	58	19.1	NaN	3
9	10	20	22.3	NaN	1
10	11	56	22.2	2653.0	5
11	12	23	29.8	2354.0	5
12	13	25	30.1	3207.0	0
13	14	44	20.5	NaN	8
14	15	26	35.0	3233.0	5

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Noisy Data

Q: What is noise?

A: Random error in a measured variable.

Incorrect attribute values may be due to

- faulty data collection instruments
- data entry problems
- data transmission problems
- technology limitation
- inconsistency in naming convention

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How to Handle Noisy Data?

If the entire dataset contains lots of noisy data, then use Binning method:

- first sort data and partition into (equi-depth) bins
- then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- used also for discretization (discussed later)

If the dataset contains some outlier noisy data, then:

Clustering: detect and remove outliers

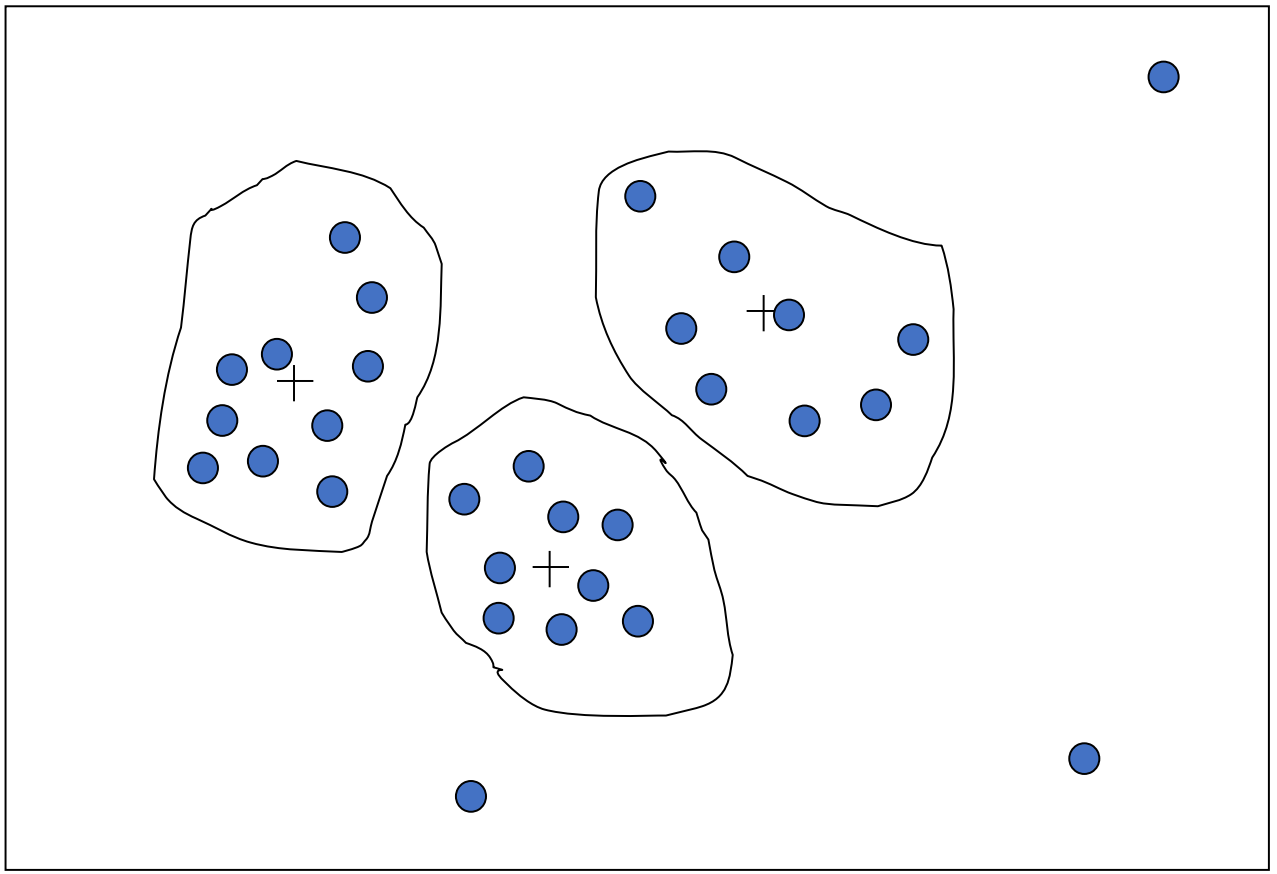
Semi-automated method: combined computer and human inspection that detects suspicious values and check manually

Regression: smooth by fitting the data into regression functions

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Cluster Analysis

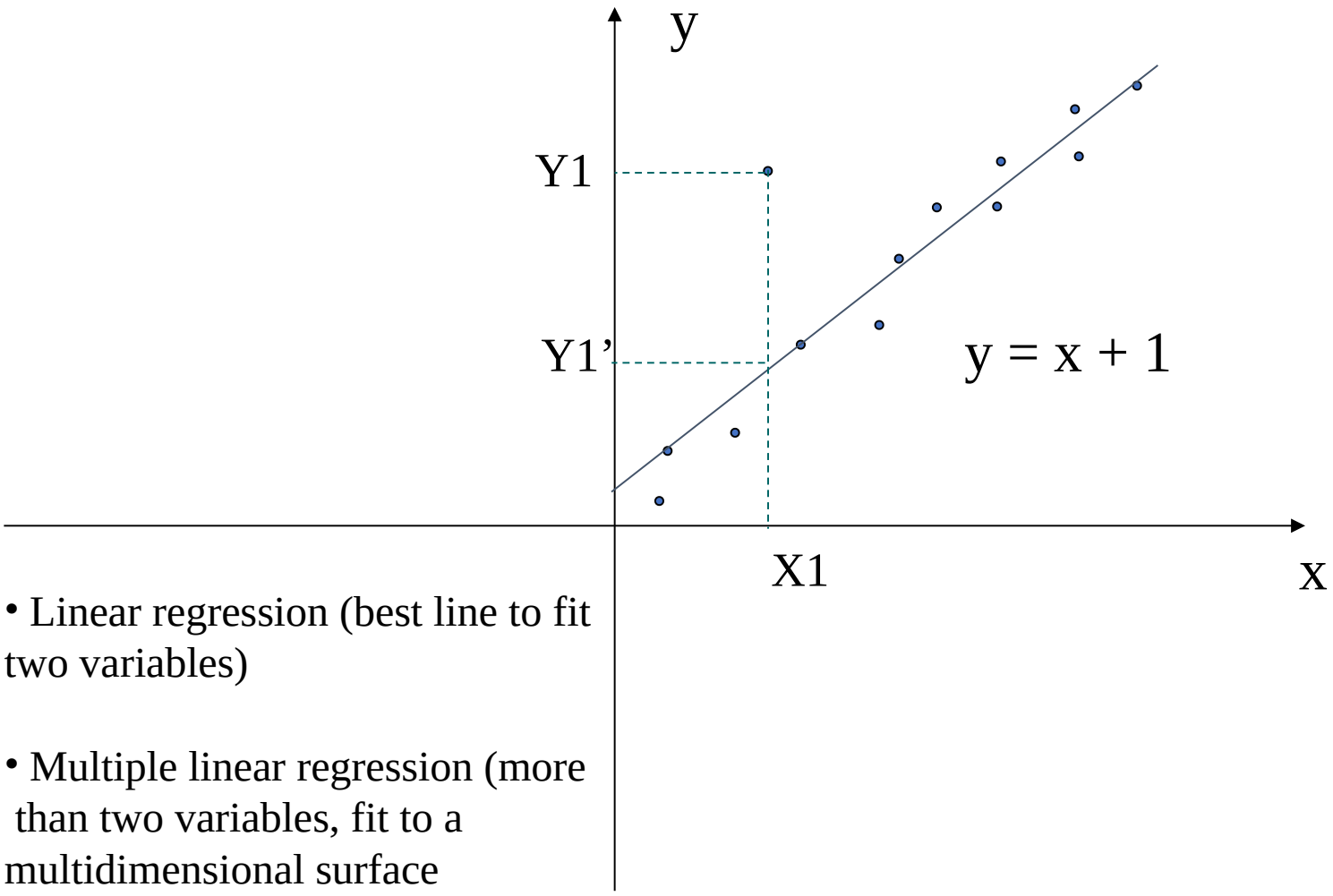


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Regression



• Linear regression (best line to fit two variables)

• Multiple linear regression (more than two variables, fit to a multidimensional surface

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How to Handle Inconsistent Data?

Manual correction using external references

Semi-automatic using various tools

- To detect violation of known functional dependencies and data constraints
- To correct redundant data

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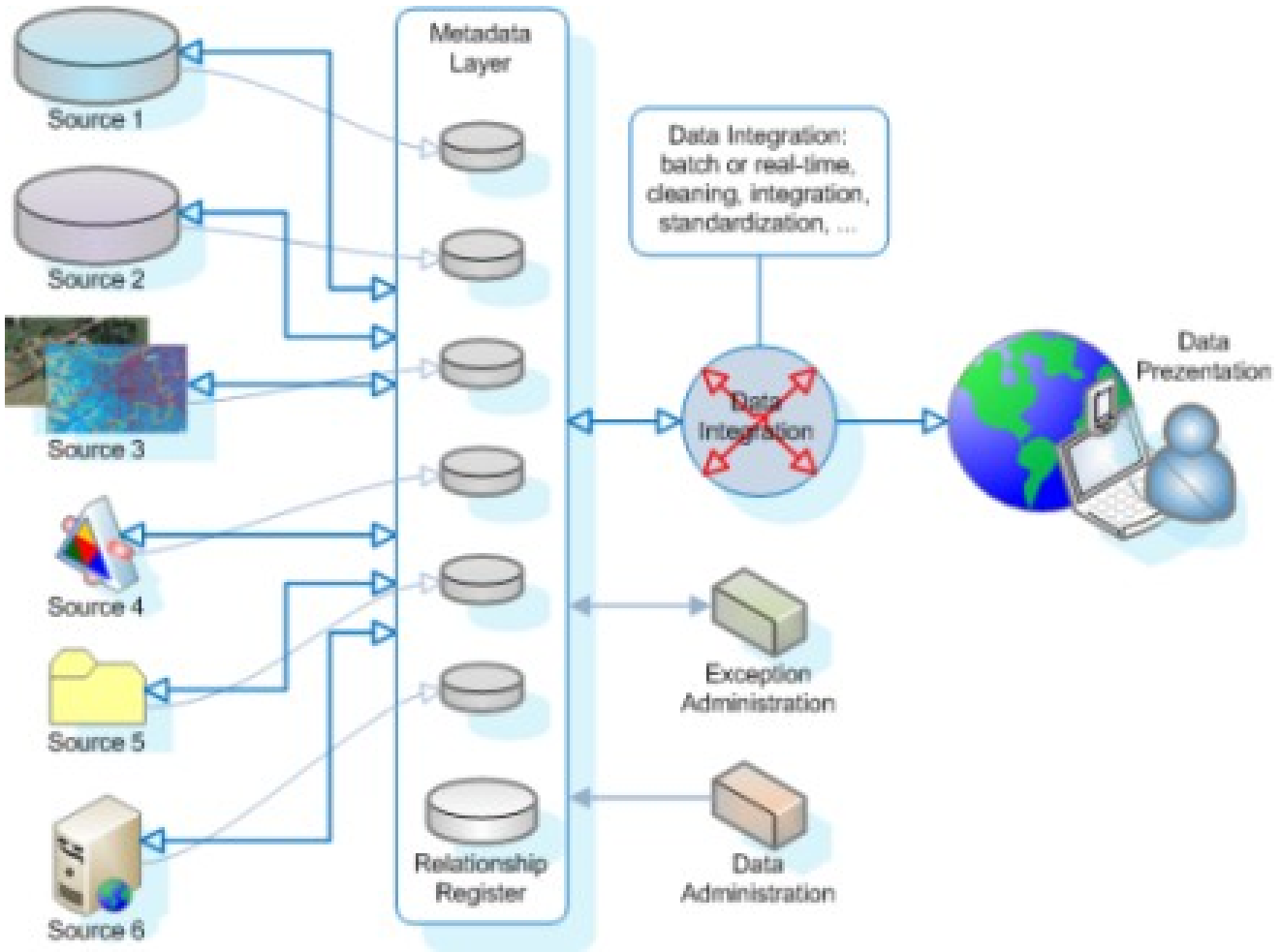
Data reduction

Summary

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Data Integration Challenges



Multipole sources

Metadata Management

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Data Integration

Data integration:

- combines data from multiple sources into a coherent store

Schema integration

- integrate metadata from different sources
- Entity identification problem: identify real world entities from multiple data sources, e.g., A.cust-id ≡ B.cust-#

Detecting and resolving data value conflicts

- for the same real world entity, attribute values from different sources are different
- possible reasons: different representations, different scales, e.g., metric vs. British units, different currency

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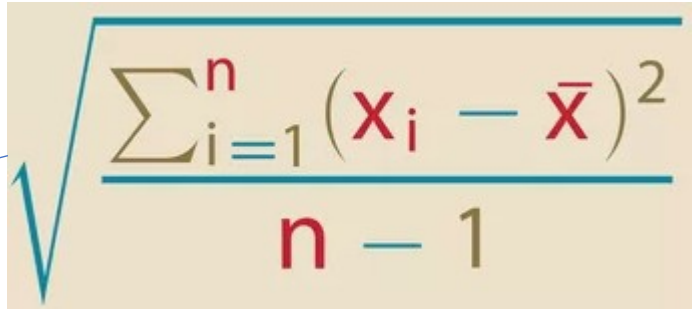
Handling Redundant Data in Data Integration

Redundant data occur often when integrating multiple DBs

- The same attribute may have different names in different databases
- One attribute may be a “derived” attribute in another table, e.g., annual revenue

Redundant data may be able to be detected by correlational analysis

$$r_{A,B} = \frac{\Sigma(A - \bar{A})(B - \bar{B})}{(n - 1)\sigma_A \sigma_B}$$



Careful integration can help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

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Data Transformation

Normalization: scaled to fall within a small, specified range

- min-max normalization
- z-score normalization
- normalization by decimal scaling

Attribute/feature construction

- New attributes constructed from the given ones

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Data Transformation: Normalization

Particularly useful for classification (NNs, distance measurements, nn classification, etc)

min-max normalization

$$v' = \frac{v - min_A}{max_A - min_A}(new_max_A - new_min_A) + new_min_A$$

z-score normalization

$$v' = \frac{v - mean_A}{stand_dev_A}$$

normalization by decimal scaling

$$v' = \frac{v}{10^j}$$

Where j is the smallest integer such that Max($|v'|$)<1

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Data Reduction

Problem:

Big Data may store terabytes of data: Complex data analysis/mining may take a very long time to run on the complete data set

Solution?

- Data reduction…

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Data Reduction

- Obtains a reduced representation of the data set that is much smaller in volume but yet produces the same (or almost the same) analytical results
- Data reduction strategies
  - Data cube aggregation
  - Dimensionality reduction
  - Data compression
  - Numerosity reduction
  - Discretization and concept hierarchy generation

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Data Cube Aggregation

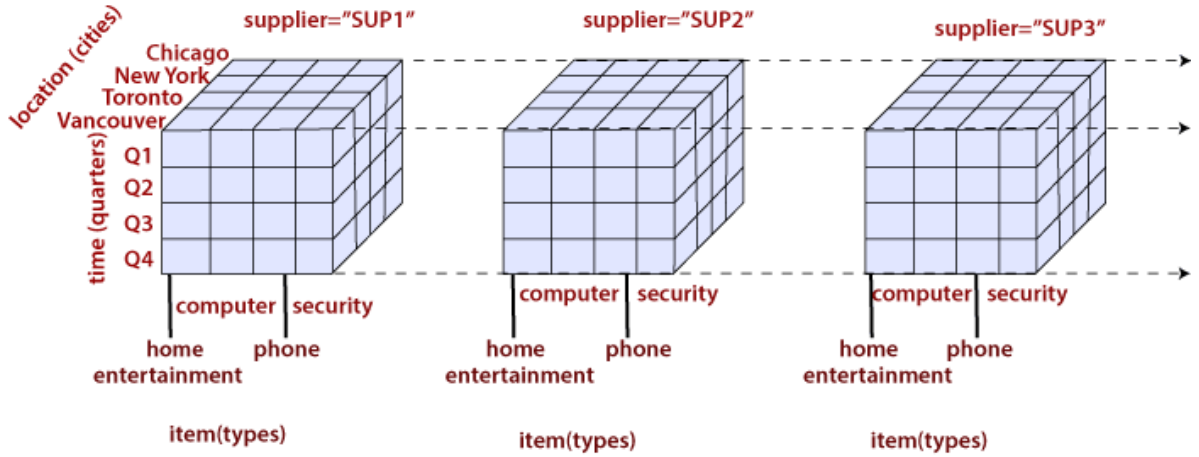
Multiple levels of aggregation in data cubes

- Further reduce the size of data to deal with

Reference appropriate levels

- Use the smallest representation capable to solve the task

Queries regarding aggregated information should be answered using data cube, when possible



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Dimensionality Reduction

Problem: Feature selection (i.e., attribute subset selection):

- Select a minimum set of features such that the probability distribution of different classes given the values for those features is as close as possible to the original distribution given the values of all features
- Nice side-effect: reduces # of attributes in the discovered patterns (which are now easier to understand)

Solution: Heuristic methods (due to exponential # of choices) usually greedy:

- step-wise forward selection
- step-wise backward elimination
- combining forward selection and backward elimination
- decision-tree induction

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Data Compression

String compression

- There are extensive theories and well-tuned algorithms
- Typically lossless
- But only limited manipulation is possible without expansion

Audio/video, image compression

- Typically lossy compression, with progressive refinement
- Sometimes small fragments of signal can be reconstructed without reconstructing the whole

Time sequence is not audio

- Typically short and vary slowly with time

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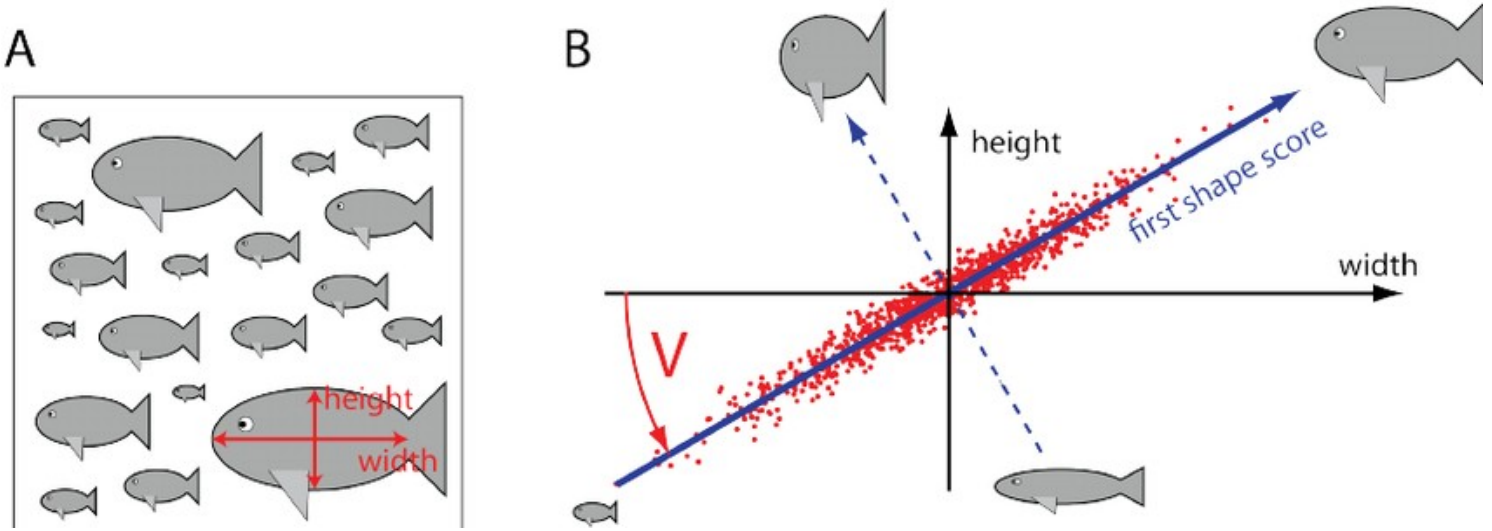
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Principal Component Analysis

PCA

•The principal components (new set of axes) give important information about variance.

• Using the strongest components one can reconstruct a good approximation of the original signal.



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Numerosity Reduction

Reduce data volume by choosing alternative, smaller forms of data representation

Parametric methods

- Assume the data fits some model, estimate model parameters, store only the parameters, and discard the data (except possible outliers)

Non-parametric methods

- Do not assume models
- Major families: histograms, clustering, sampling

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Sampling

Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data

Cost of sampling: proportional to the size of the sample, increases linearly with the number of dimensions

Choose a representative subset of the data

- Simple random sampling may have very poor performance in the presence of skew

Develop adaptive sampling methods

- Stratified sampling:
  - Approximate the percentage of each class (or subpopulation of interest) in the overall database
  - Used in conjunction with skewed data

Sampling may not reduce database I/Os (page at a time).

Sampling: natural choice for progressive refinement of a reduced data set.

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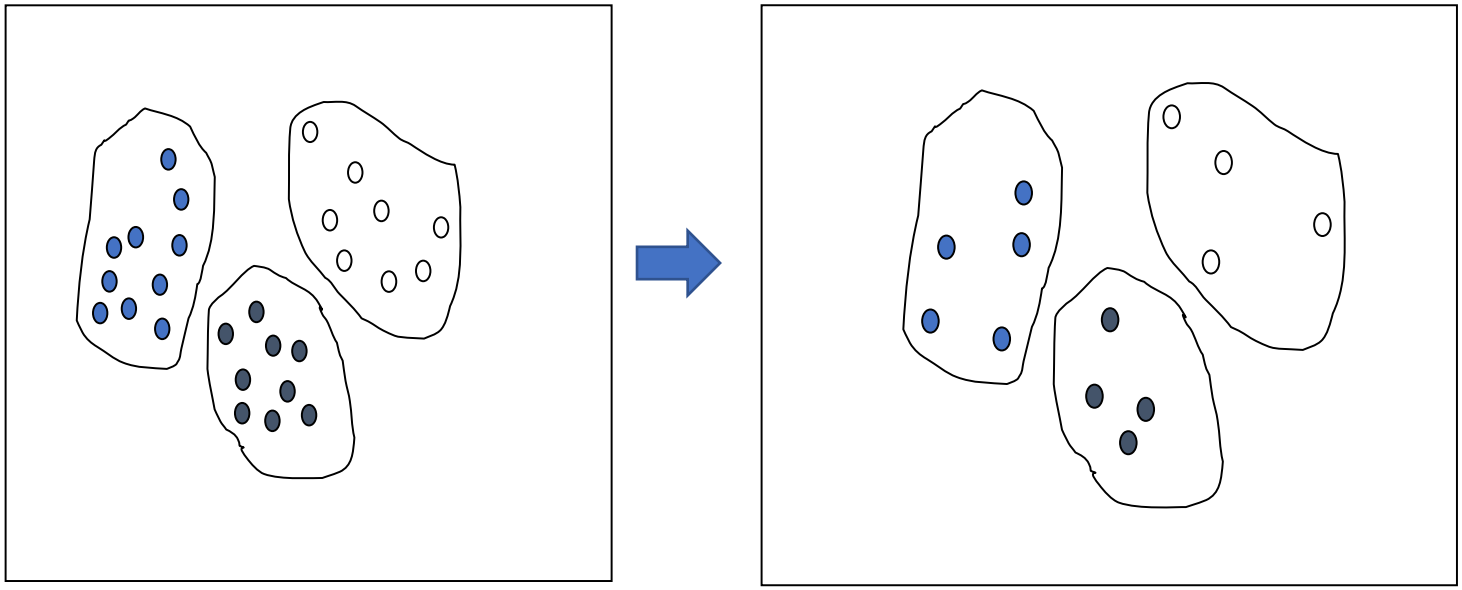
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Sampling

Raw Data

Cluster/Stratified Sample



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Summary

Data preparation is a big issue for data mining but the important steps for not very big data are handle missing data, noisy data and feature selection (plus EDA)

A lot a methods have been developed in data pre-processing but still an active area of research

After Break (10 minutes)

Lecture: EDA (Exploring Data Analysis)

Practical: Data pre-procssing and EDA

Homework

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